CSCI-3171 Network Computing
Lab 3 – Winter 2026

Submission Instructions:

Submit **one PDF file** for your lab report on Brightspace under the Lab 3 portal. Include the following:

1. Summary Report: Provide a summary of your observations for the given questions.
2. Graphs and Numerical Values: Include any relevant graphs and numerical values derived from your experiments.
3. Screenshots: If necessary, include screenshots. However, note that command responses can be easily copy-pasted into the report without the need for heavy screenshots.
4. Identification: Ensure your full name and banner ID are clearly mentioned in the report.

Late submission: Refer to the syllabus on Brightspace for a complete description of the late submission policy.

The Internet architecture operates in **layers**. Imagine you want to send the text message "Hello, how are you?" to a friend. Your smartphone's messaging **application** (e.g., WhatsApp) breaks down your text message into smaller units called packets. The **transport layer** (e.g., TCP or UDP) adds a header to each packet, specifying the source and destination port numbers, as well as other control information. The **network layer** (e.g., IP) adds another header to each packet, including the source and destination IP addresses. This header helps routers on the internet route the packets to their destination. The **data link layer** (e.g., Ethernet) adds a final header to each packet, containing MAC addresses that identify the physical devices (e.g., routers, switches) on the local network. Finally, the encapsulated packets are then transmitted over the internet, hopping from router to router until they reach the destination. Upon reaching the destination network, the headers are processed by the layers in reverse order, starting from the physical layer. The headers are successively removed as the packets move up the network stack. The recipient's messaging application then reassembles the received packets, reconstructing the original text message "Hello, how are you?" for the recipient to read.

1. HTTP messages

Now, let us utilize Wireshark to delve into the various headers of packets involved in web communication. Start by downloading the packet capture file **`lab3-capture.pcapng`** and open it with Wireshark. Simply double-click on the file to launch Wireshark and start analyzing the captured packets.

Recall from Lab 2 the various elements in the Wireshark graphical interface depicted in Figure 1. Let us identify these elements and their corresponding names as we will reference them in the following discussion:

- Command menu
- Display filter specification
- Listing of captured packets
- Details of selected packets
- Packet contents

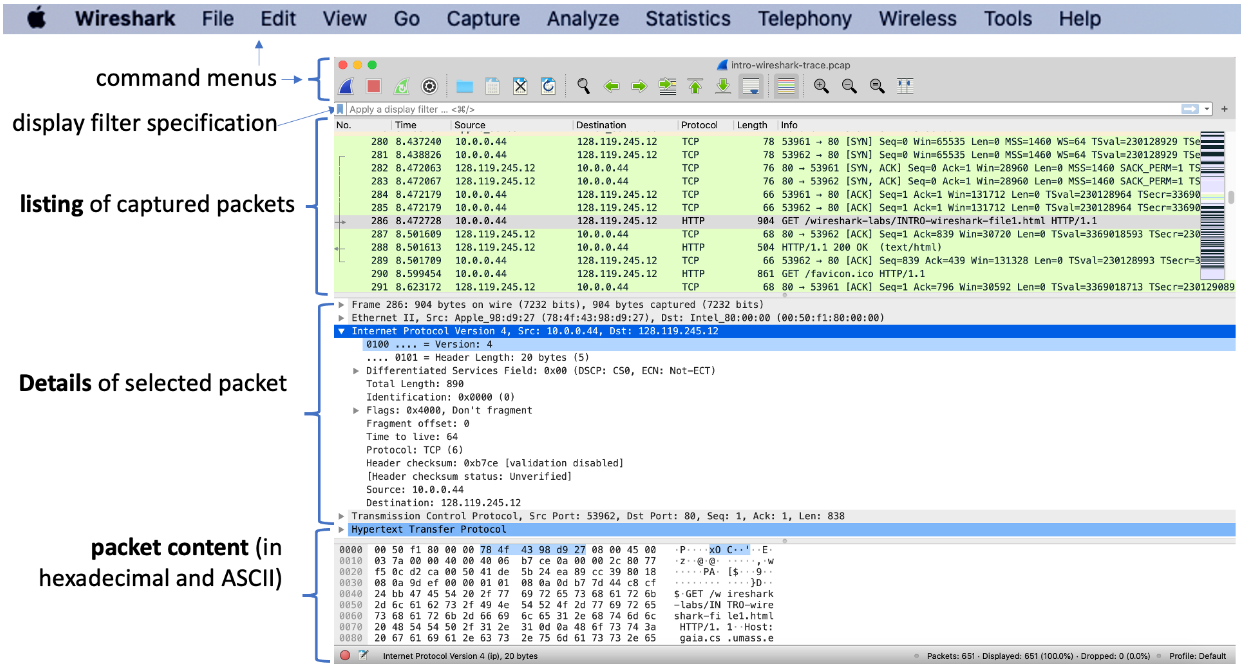

Figure 1 - Wireshark graphical interface.

Right underneath the command menu, you can specify a filter that you want to apply to the packets that you see. Type in **`http`** (without the quotes, and *in lower case* – all protocol names are in lower case in Wireshark) into the display filter specification window at the top of the main Wireshark window. Then select *Apply* (to the right of where you entered your filter) or just hit return. This will cause only HTTP message to be displayed in the packet-listing window. Figure 2 below shows a screenshot after the http filter has been applied.

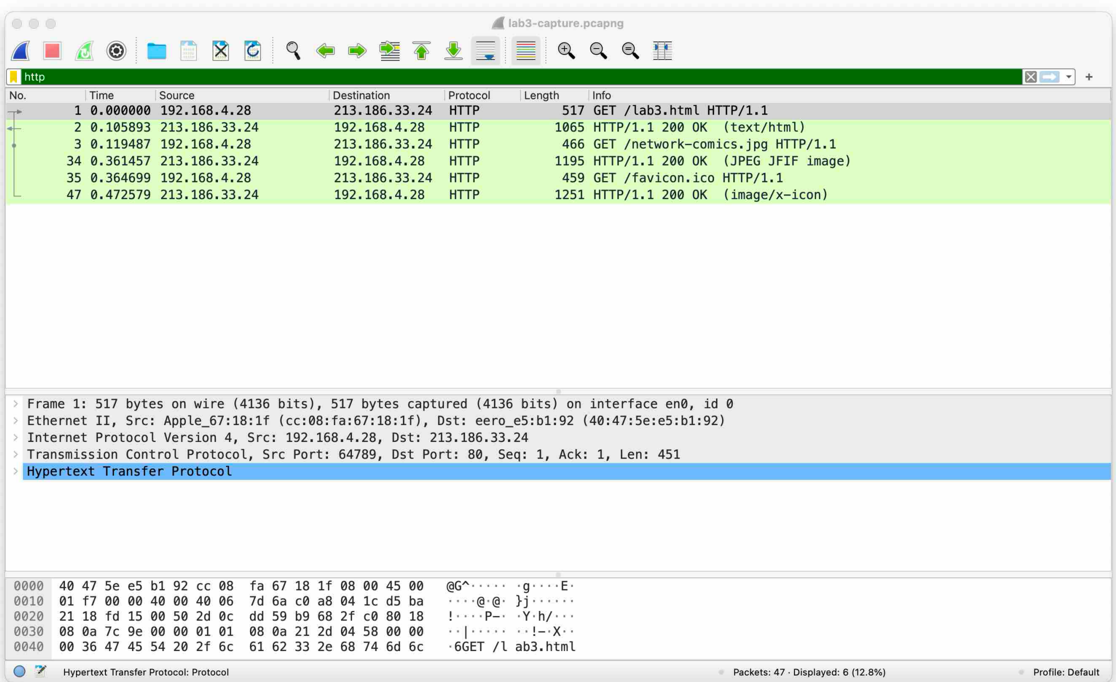

Figure 2 - http filter applied on packet capture file.

Now answer the questions below. You should **use screenshots** when appropriate to support your answers.

Q1.	How many packets were captured in the HTTP exchange? (1 point)
Q2.	This communication involves two end-hosts. Their IP addresses appear in the Source and Destination fields in the listing of captured packets. What are the source and destination IP addresses involved in the HTTP communication? (1 point)
Q3.	How many packets are being sent and received by each end-host? (1 point)

In the listing of captured packets, the info field shows basic information about each message. Particularly, you can see what object is being requested in each HTTP message using the **`GET`** command.

Q4.	What object is being requested in packet number 1. (1 point)

Select packet number 1 in the listing. Unfold the HTTP header in the details of selected packet window and locate the field **`Request URI`.**

Q5.	Can you identify which URL the user has typed in the browser? (1 point)
Q6.	Can you determine the IP address of the HTTP client (browser) and the IP address of the server involved in the communication? How did you proceed? (2 points)
Q7.	What is the status code returned in the HTTP response in packet number 2? What does this status code indicate? (2 points)

Select packet number 2 in the listing. Unfold all the HTTP headers in the details of selected packet window.

Q8.	What is the content of **`Line-based text data`**? (1 point)
Q9.	Can you summarize the exchange in packets number 1 and 2? (1 point)

Select packet number 3 in the listing.

Q10.	Who is sending the packet? (1 point)
Q11.	What is the object being requested? (1 point)

Select packet number 34 in the listing.

Q12.	Who is sending the packet? (1 point)
Q13.	What is the object being returned? (1 point)

For packet number 34, unfold all the HTTP headers in the details of selected packet window and locate **`JPEG File Interchange Format`**. Right click on **`JPEG File Interchange Format`** and select **`Show Packet Bytes…`**.

Q14.	What can you observe in the details of the packet bytes? (1 point)
Q15.	Can you comment on why using Wireshark and capturing packets could potentially pose a security risk? (1 point)
Q16.	For packets number 35 and 47, what is the object requested/returned? How is this object typically used in modern browsers? (2 points)

Now click on `**Statistics**` in the top menu then `**Flow Graph**` and then check `**Limit to display filter**`.

Q17. Join a screenshot of the flow graph window and comment the result. (2 points)

2. Packet Encapsulation

In the following, our focus will be on packet number 1. We will analyze the encapsulation and examine the content of the various headers.

Select packet number 1 and search for the answers to the following questions in the details of the selected packet. Keep the headers folded as in Figure 2 for a start and then unfold them if needed. The top line in the details is the frame captured on the physical layer. The following lines are the headers of the upper layers.

Q18. How many layers on top of the physical layer can you identify in this packet capture? (1 point)
Q19. What is the length of the captured packet on wire? (1 point)

Q20. What is the protocol used at the link layer? (1 point)
Q21. What is the size of the link layer header? Hint: you can select the link layer header in the packet detail and look for the bottom-left corner of the Wireshark window, as illustrated in Figure 3. (1 point)
Q22. What is the source MAC address used in the link layer header? What information can we get from the MAC address about the end-host? (2 points)

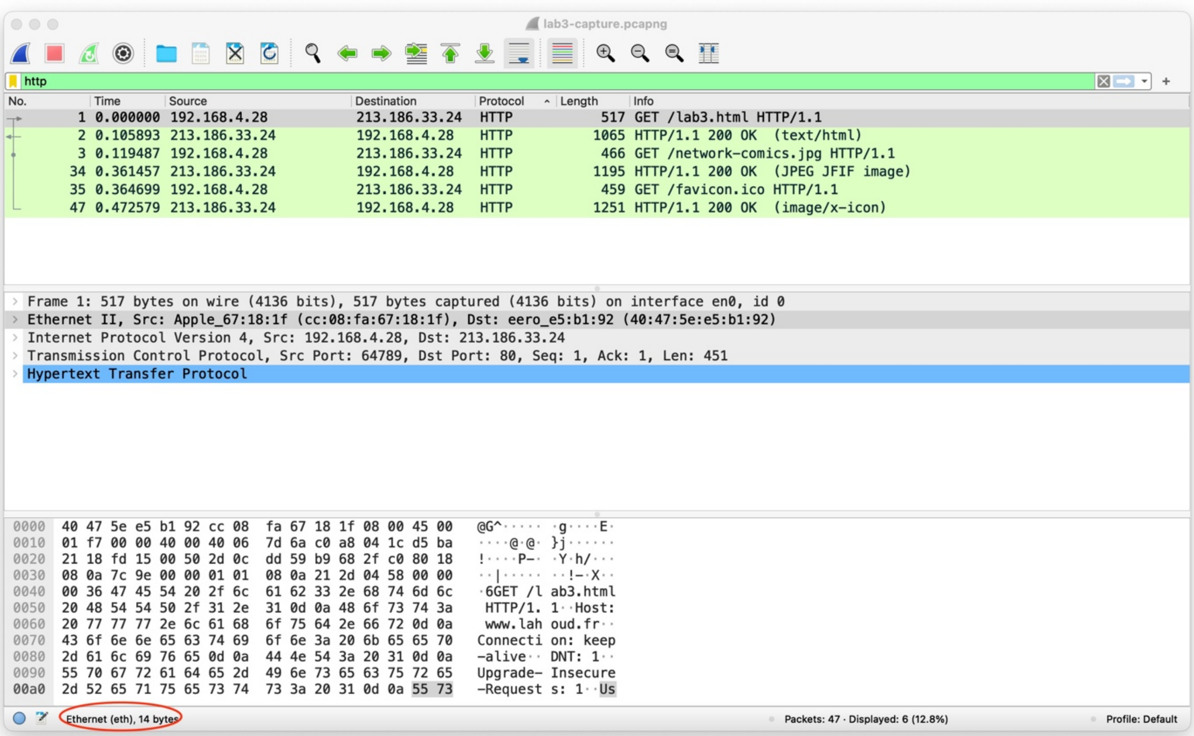

Figure 3 - Header size for link layer.

Q23.	What is the protocol used at the network layer? (1 point)
Q24.	What is the size of the network layer header? (1 point)
Q25.	What is the destination IP address used in the network layer header? Can you determine which web server is associated with this address? (2 points)

Q26.	What is the protocol used at the transport layer? (1 point)
Q27.	What is the size of the transport layer header? (1 point)
Q28.	What is the destination and source ports in the transport layers header? Why are ports necessary for identifying processes on the Internet? (2 points)

Q29.	What is the protocol used at the application layer? (1 point)
Q30.	What is the size of the message sent by the application? (1 point)

Considering that the message size specified by the application header is useful information, while all headers by lower layers are considered overhead.

Q31.	What is the total overhead (computed as a ratio) in packet number 1? (1 point)
Q32.	What is the overhead added by each layer (transport, network, and link) in packet number 1? Provide your detailed observations. (2 points)